

A PRIMER ON NONLINEAR CONTROL WITH APPLICATIONS IN SYNTHETIC BIOLOGY

VIPUL SINGHAL

ABSTRACT. We present a brief overview of nonlinear observability, controllability and identifiability, and outline applications in systems and synthetic biology. We begin with some foundational ideas from geometry, namely manifolds, vector and covector fields, Lie derivatives and brackets, and Frobenius' theorem), followed by key tests of controllability and observability, and finally describe minimal realizations of nonlinear systems. This leads to practical methods for identifying unobservable or unreachable states, along with non-identifiable parameters in nonlinear systems.

1. INTRODUCTION

Differential geometry provides the mathematical foundation for understanding and controlling nonlinear systems, a crucial capability in modern engineering and science. While linear approximations offer valuable insights, many systems in engineering and science, including spacecraft attitude control, robot manipulation, and biological systems, exhibit inherently nonlinear behavior. Consider the intricate dynamics of a gene regulatory network: genes interact through complex feedback loops, proteins modify expression levels through nonlinear activation and repression functions, and epigenetic states evolve through intricate biochemical mechanisms. Unlike linear systems, where effects are proportional to inputs, these networks exhibit rich behaviors like bistability, oscillations, and threshold effects, making both control and state estimation particularly challenging.

The mathematical tools we will develop in this text – smooth manifolds, tangent spaces, vector fields, and more – provide the language to describe and analyze these nonlinear systems. These tools have enabled breakthroughs in diverse fields, including aeronautics, robotics, and machine learning. In aeronautics, they underpin the precision control of aircraft and spacecraft orientation. In robotics, they facilitate motion planning for robotic arms and autonomous vehicles. In machine learning, they provide insights into the geometry of optimization landscapes, enhancing our understanding of how algorithms converge.

This text focuses on applying these powerful tools to understand and control genetic systems. For example, we can use differential geometry to design control strategies for steering cellular states through precise combinations of drug dosing and optogenetic interventions. Imagine controlling the expression of a specific gene involved in cancer progression by manipulating the activity of key regulatory proteins.

Two fundamental concepts we'll explore in depth are nonlinear observability and controllability. Observability addresses whether we can infer a system's full state – such as protein concentrations or epigenetic modifications – from partial measurements, like RNA sequencing or fluorescent reporter data. Controllability examines whether we can drive the system to desired states using available interventions. In the context of our example, this might involve determining whether we can effectively control the expression of the cancer-related gene by manipulating the activity of its regulators.

A closely related concept, parameter identifiability, focuses on determining which kinetic parameters in a model can be inferred from experimental data – a critical step in building accurate representations of cellular processes. These concepts form the foundation for understanding and controlling the intricate behavior of biological systems.

This text begins with fundamental concepts from differential geometry, including smooth manifolds, coordinate charts, tangent spaces, and flows generated by vector fields. These concepts form the foundation for analyzing key properties such as controllability, observability, and identifiability in biological systems. For a deeper dive into these concepts, readers are encouraged to consult [Nijmeijer and Van Der Schaft(1990)] for

an excellent introduction to control applications and [Boothby(2002)] for a thorough grounding in geometric foundations. For a more advanced perspective, [Abraham et al.(1988)] Abraham, Marsden, and Ratiu deliver an in-depth treatment of these ideas.

This text aims to equip readers with the theoretical understanding and practical tools to analyze and control complex biological systems, ultimately paving the way for advancements in areas such as personalized medicine and synthetic biology.

2. DIFFERENTIAL GEOMETRIC PRELIMINARIES

2.1. Notational conventions. We adopt the Einstein summation convention throughout: repeated indices (once as superscript, once as subscript) imply summation over their range. For example, given a vector v and basis vectors $\{E_1, \dots, E_n\}$, we have

$$(1) \quad v = \alpha^i E_i \triangleq \sum_{i=1}^n \alpha^i E_i$$

All vector spaces will be real and finite-dimensional, and vectors and their coefficients will be indexed with subscripts and superscripts, respectively (as in Eq (1)). Similarly, covectors will have superscripts, while the corresponding coefficients will have subscripts, $\omega = \sum_{i=1}^n \omega_i dx^i = \omega_i dx^i$. Furthermore, the index i in $\frac{\partial}{\partial x^i}$ will be interpreted as a subscript; indeed, we will regularly abbreviate $\frac{\partial}{\partial x^i}|_a$ to ∂_{ia} , and later see that this is an element of $T_a \mathbb{R}^n$, the tangent space to $\mathbb{R}^n$ at a point a . Note also that all vector spaces in this document will be finite dimensional.

2.2. Manifolds, charts and local representatives. A smooth manifold is a set which is locally Euclidean—in the sense that neighborhoods of points can be modeled using Euclidean spaces—along with a certain *differentiable structure*, which we describe below. We use the symbol N to denote an n -dimensional smooth (C^∞) manifold, and the tuples (U, φ) and (V, ϕ) to denote coordinate charts, where U and V are open sets in N and the homeomorphisms φ and ϕ map them (respectively) onto open sets in $\mathbb{R}^n$. The key component of N 's differentiable structure is the following compatibility requirement for coordinate changes: whenever $U \cap V$ has a nonempty interior, the map $\phi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \phi(U \cap V)$ is a (smooth) diffeomorphism.

Remark 2.1. For simplicity, we omit certain technical requirements on the topology of N , such as the existence of a second countable basis of open sets and the Hausdorff property.¹

A point $p \in U$ in local coordinates will typically be denoted by $a = \varphi(p) = (x^1(p), \dots, x^n(p)) \in \mathbb{R}^n$, where the $x^i : U \rightarrow \mathbb{R}$ are the (C^∞) *coordinate functions* corresponding to a given φ . Coordinate neighborhoods may therefore also be denoted by $(U, x^1, \dots, x^n)$.

The local representative of a map $f : U \rightarrow \mathbb{R}$ is constructed as $\hat{f} \triangleq f \circ \varphi^{-1}$, so that

$$(2) \quad f(p) = \hat{f}(\varphi(p)) = \hat{f}(a) = \hat{f}(x^1(p), \dots, x^n(p)).$$

Per convention, we simplify notation by dropping the caret from f and the explicit dependence of the coordinate functions on p , resulting in the last two expressions in Equation 2 being written $f(a) = f(x^1, \dots, x^n)$.

2.3. The Inverse and Implicit Function Theorems, and the Rank Theorem. In this section, we describe the inverse function theorem, which is one of the foundational results in calculus. We also describe two corollaries—the implicit function and rank theorems—which play an instrumental role in differential geometry.

Consider two manifolds N_1 and N_2 of dimension n_1 and n_2 respectively, with respective coordinate charts (U, φ) and (V, ϕ) around points $p \in U \subset N_1$ and $F(p) \in V \subset N_2$, where $F : N_1 \rightarrow N_2$ is a map between the manifolds. Its local representative is given by $\hat{F} : \mathbb{R}^{n_1} \rightarrow \mathbb{R}^{n_2}$, defined by $(x^1, \dots, x_{n_1}) \mapsto \phi \circ F \circ \varphi^{-1}(x^1, \dots, x_{n_1})$. As usual, we will drop the caret, and overload F by using it to refer to both the map between manifolds, and its local representative, since its meaning will be clear from context. F is called a *diffeomorphism* if it is bijective and both (the local representatives of) F and F^{-1} are smooth; if such a diffeomorphism exists, then (i) $n_1 = n_2$ and (ii) N_1 and N_2 are called diffeomorphic.

¹These properties ensure that the manifold is well-behaved and has a countable atlas of charts. While important for a complete understanding, they can be safely assumed for most applications in nonlinear control. For a thorough treatment, see Chapter 1 of [Abraham et al.(1988)] Abraham, Marsden, and Ratiu.

- describe the three theorems
- can also briefly mention the few propositions (without proof.)
- and its local representative
- submersion
- immersion

2.4. Vectors as differential operators and the pushforward map. In this section, we define the tangent space to a manifold, vectors as differential operators on real-valued functions, and the idea of the pushforward of a vector.

2.4.1. Tangent Space to $\mathbb{R}^n$. We begin with defining the *tangent space* to $\mathbb{R}^n$ at p , denoted by $T_p\mathbb{R}^n$, in a few steps. Denote by $C^\infty(p)$ the set of all real-valued smooth functions whose domain of definition contains a neighbourhood of $p \in \mathbb{R}^n$, identifying those functions that agree on an open set containing p . Next, let $T_p\mathbb{R}^n$ be an n -dimensional vector space, with basis $\{E_{ip}\}_{i=1}^n$, and its elements, called the *tangent vectors* at p , written as linear combinations of the form (via the summation convention),

$$(3) \quad X_p = \alpha^i E_{ip},$$

with coefficients $\alpha^i \in \mathbb{R}$. Tangent vectors of the form 3 will be identified with (not necessarily unit-norm) directional derivatives of elements of $C^\infty(p)$, defined as

$$(4) \quad X_p(f) \triangleq \sum_{i=1}^n \alpha^i \left(\frac{\partial f}{\partial x^i} \right)_p = \alpha^i \partial_{ip} f,$$

where the subscript p denotes the point at which the derivative is evaluated.

Interpreting X_p as a directional derivative means that it can be thought of as a map from $C^\infty(p)$ to $\mathbb{R}$. This motivates the definition of a particular type of operator on real-valued smooth functions, called a *derivation*.

Definition 2.2 (Derivation). The space of *derivations*, denoted $\mathcal{D}(p)$, is the vector space of mappings $C^\infty(p) \rightarrow \mathbb{R}$ such that for all $\alpha, \beta \in \mathbb{R}$ and $f, g \in C^\infty(p)$ the following two properties,

$$(5) \quad \begin{aligned} X_p(\alpha f + \beta g) &= \alpha(X_p f) + \beta(X_p g) && \text{(linearity),} \\ X_p(fg) &= (X_p f)g(p) + f(p)(X_p g) && \text{(Leibniz rule),} \end{aligned}$$

hold, with vector space operations,

$$(6) \quad \begin{aligned} (X_p + Y_p)(f) &= X_p(f) + Y_p(f), \\ (\alpha X_p)(f) &= \alpha X_p(f), \quad \alpha \in \mathbb{R}. \end{aligned}$$

It may be verified that directional derivatives satisfy these properties, and are therefore derivations (and in fact, on the algebra of smooth functions $C^\infty(p)$, these are known to be the *only* type of derivations to exist.)² These ideas are tied together by the following theorem.

Theorem 2.3. *The tangent space $T_p\mathbb{R}^n$ is isomorphic to the vector space $\mathcal{D}(p)$ of all derivations of $C^\infty(p)$ into $\mathbb{R}$. This isomorphism is given by making each X_p correspond to the directional derivative in the direction of X_p .*

Using this isomorphism, we can identify $T_p\mathbb{R}^n$ with $\mathcal{D}(p)$. We write vectors as linear combinations of basis vectors, $X_p = \alpha^i \partial_{ip}$, so that the E_{ip} are identified with ∂_{ip} , and $E_{ip}f = \partial_{ip}f$. Note that $X_p x^i = (\alpha^j \partial_{jp}) x^i = \alpha^j \delta_j^i = \alpha^i$, because $\frac{\partial x^i}{\partial x^j} = \delta_j^i = 1$ if $i = j$ and 0 otherwise (δ_j^i is called the Kronecker delta). Consequently, X_p is completely determined by its action on the coordinate functions, x^i . This equivalence between derivations on functions and the tangent space will be especially useful in defining the tangent space to manifolds, a topic we cover next.

²which is not the case on $C^r(p)$, for $r < \infty$

2.4.2. *Tangent Space to N .* Consider a smooth manifold N and a point $p \in N$. We may define $C^\infty(p)$ as the algebra of real-valued smooth functions whose domain of definition contains some neighbourhood of p in N , with functions identified if they agree on any neighbourhood of p .

Definition 2.4 (Tangent Space to N). The *tangent space* to N at point p , denoted by $T_p N$, is defined as the space of derivations on $C^\infty(p)$, that is, the vector space of mappings $X_p : C^\infty(p) \rightarrow \mathbb{R}$ satisfying the linearity and Leibniz rule properties defined in Equation 5. Elements of this space, denoted by $X_p \in T_p N$, are called *tangent vectors* at p .

2.4.3. *Tangent Bundle of N .* The tangent bundle associated with a manifold is defined as the disjoint union of individual tangent spaces (that is, keeping track of the base points in addition to the tangent spaces themselves),

$$\begin{aligned} TN &= \bigsqcup_{p \in N} T_p N \\ (7) \quad &= \bigcup_{p \in N} \{p\} \times T_p N \\ &= \bigcup_{p \in N} \{(p, v) \mid v \in T_p N\}. \end{aligned}$$

One also defines the natural projection $\pi : TN \rightarrow N : (p, v) \mapsto p$. Note that some authors define the tangent bundle without the base point explicitly included, $TN = \cup_{p \in N} T_p N$, with projection $v \mapsto \pi(v) = p$ for $v \in T_p N$.

In some cases, it is possible to find an isomorphism between the tangent bundle and the Cartesian product of the base space and some vector space, as in the cases of $T\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n$ or $T\mathbb{S}^1 \cong \mathbb{S}^1 \times \mathbb{R}$. In this case, the tangent bundle is called the *trivial bundle*. In general, however, such a decomposition will not exist (the most common example being $T\mathbb{S}^2$), and we will always use the definition in equation 7. The tangent bundle plays an instrumental role in the definition of (smooth) vector fields on manifolds (Section 2.5 below).

2.4.4. *Pushforward (differential) of a map.* To describe elements of $T_p N$ in local coordinates, we require two definitions.

Definition 2.5 (Pullback and Pushforward). Let $F : N \rightarrow M$ be a C^∞ map of manifolds. Then, for $p \in N$ and $f \in C^\infty(F(p))$, the *pullback* $F^* : C^\infty(F(p)) \rightarrow C^\infty(p)$ defined by $F^*(f) = f \circ F$ is a homomorphism of algebras. Its dual vector space homomorphism $F_* : T_p N \rightarrow T_{F(p)} M$, is called the *pushforward*, and is defined by $F_*(X_p)f = X_p(F^*f)$, where $F_*(X_p)$ is a map of $C^\infty(F(p))$ to $\mathbb{R}$. If $H = G \circ F$, then $H^* = F^* \circ G^*$ and $H_* = G_* \circ F_*$. If F is the identity, then F^* and F_* are the identity isomorphisms (on their respective domains).

A corollary of the last statement in Definition 2.5 is that pushforwards created out of diffeomorphisms are isomorphisms (consider, for instance, the identity isomorphism $F_*^{-1} \circ F_*$ constructed out of the identity map $F^{-1} \circ F$). An important implication of this is that, for a chart $(U \ni p, \varphi)$, the spaces $T_p U$ and $T_a(\varphi(U)) \cong T_a \mathbb{R}^n$ are isomorphic, with $a = \varphi(p)$. The basis $\{\partial_{ia}\}_{i=1}^n$ of $T_a \mathbb{R}^n$ defines a basis of $T_p U$,

$$\begin{aligned} (8) \quad &\varphi_*^{-1} : T_a \mathbb{R}^n \rightarrow T_p U, \quad a = \varphi(p), \\ &E_{ip} \triangleq \varphi_*^{-1} \partial_{ia}, \quad i = 1, \dots, n. \end{aligned}$$

This basis is called a coordinate frame at $p \in U$, and the coordinate frames on U comprise the collection of such bases on all points in U , that is, $\{\{E_{ip}\}_{i=1}^n \mid p \in U\} \subset TU$. When referring to the frames on U as a whole, or to an arbitrary frame (with base point p unspecified), we will use the symbols $E_1, \dots, E_n$, or, in local coordinates, $\partial_1, \dots, \partial_n$. Notice that unlike in the $T_p \mathbb{R}^n$ case discussed previously, $\{E_i\}_{i=1}^n$ and $\{\partial_i\}_{i=1}^n$ cannot be identified with each other, although are equivalent up to isomorphism (via φ_*), which reduces the study of $T_p U$ to that of $T_a \mathbb{R}^n$.

Given $f \in C^\infty(p)$ with local representative $\hat{f} = f \circ \varphi^{-1}$, we have $E_{ip}f = \partial_{ia}\hat{f}$. In particular, for coordinate functions x^j , we have $E_{ip}x^j = \partial_i \hat{x}^j = \delta_i^j$. A vector may then be written $X_p = \alpha^i E_{ip}$, with $\alpha^i = X_p x^i$. With these definitions, we can express $X_p f$ in local coordinates,

$$(9) \quad X_p f(p) = \alpha^i E_{ip} f(p) = \alpha^i (\varphi_* E_{ip})(\varphi^{-1*} f)(\varphi(p)) = \alpha^i \left(\partial_{ia} \hat{f} \right) (a),$$

or, dropping the caret, $X_p f(p) = \alpha^i \partial_{ia} f(a)$.

Next, we give a theorem that describes the mapping $F_* : T_p N \rightarrow T_{F(p)} M$ in terms of coordinates and frames. Let $(U, \varphi) = (U, x^1, \dots, x^n)$ and $(V, \phi) = (V, y^1, \dots, y^m)$ be charts on N and M respectively, with $F(U) \subset V$, $a = \varphi(p) = (x^1, \dots, x^n)$. Let the local representative of $F(p)$ be $f(a) = \phi \circ F \circ \varphi^{-1}(x^1, \dots, x^n)$, with components given by $f^j(x^1, \dots, x^n) = y^j(F \circ \varphi^{-1}(x^1, \dots, x^n))$. Some authors use the symbols f^j and y^j interchangeably, but note that their domains are different— $\varphi(U)$ and V respectively—and we will treat them as distinct functions. Furthermore, in this section we will distinguish the local representatives of the bases of $T_p N$ and $T_{F(p)} M$ by writing them out in full: $\frac{\partial}{\partial x^i}$ and $\frac{\partial}{\partial y^j}$.

Theorem 2.6. Define $E_{ip} = \varphi_*^{-1}(\partial/\partial x^i)$ and $\tilde{E}_{jF(p)} = \phi_*^{-1}(\partial/\partial y^j)$ (over all i, j) as the bases of their respective tangent spaces. Then,

$$(10) \quad F_*(E_{ip}) = \left(\frac{\partial f^j}{\partial x^i} \right)_{\varphi(p)} \tilde{E}_{jF(p)}, \quad i = 1, \dots, n.$$

Furthermore, if $\beta^j \tilde{E}_{jF(p)} = F_*(\alpha^i E_{ip})$, then the components are related by

$$(11) \quad \beta^j = \alpha^i \frac{\partial f^j}{\partial x^i}, \quad j = 1, \dots, m.$$

Proof. We are looking for an expression for $F_*(E_{ip})$ in terms of the basis $\tilde{E}_{jF(p)}$, $j = 1, \dots, m$. Recall that the components of a vector may be obtained by its action on coordinate functions:

$$(12) \quad F_*(E_{ip})y^j = \left(F_* \circ \varphi_*^{-1} \left(\frac{\partial}{\partial x^i} \right)_{\varphi(p)} \right) y^j = \frac{\partial}{\partial x^i} \Big|_{\varphi(p)} (y^j \circ F \circ \varphi^{-1}) = \left(\frac{\partial f^j}{\partial x^i} \right)_{\varphi(p)}$$

Thus, $F_*(E_{ip}) = \left(\frac{\partial f^j}{\partial x^i} \right) \tilde{E}_{jF(p)}$. The second part follows by recalling the linearity of F_* and collecting terms, and is left as an exercise for the reader. $\square$

In matrix terms, Eq. 11 can be written

$$(13) \quad \begin{bmatrix} \beta^1 \\ \vdots \\ \beta^m \end{bmatrix} = \begin{bmatrix} \frac{\partial f^1}{\partial x^1} & \dots & \frac{\partial f^1}{\partial x^n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f^m}{\partial x^1} & \dots & \frac{\partial f^m}{\partial x^n} \end{bmatrix} \begin{bmatrix} \alpha^1 \\ \vdots \\ \alpha^n \end{bmatrix},$$

with the terms $\frac{\partial f^j}{\partial x^i}$ forming the *Jacobian matrix*. F_* is also known as the *differential* of F , denoted by dF .

2.4.5. The tangent velocity vector. The pushforward allows us to define the notion of the tangent *velocity vector* to a curve. Roughly speaking, given a curve $F : (a, b) \rightarrow N$, the pushforward of the vector $\partial_{t,t_0} \triangleq \frac{d}{dt} \Big|_{t_0} \in T_{t_0}(a, b)$ via F_* results in the velocity vector $F_*(\partial_{t,t_0}) \in T_{F(t_0)} N$; we can think of this as “differentiating” F with respect to t (at t_0). We describe this more formally next.

Consider the manifolds $M = (a, b)$ and N , along with a map $F : M \rightarrow N$, and note that ∂_{t,t_0} is a basis for M at t_0 . Let $p = F(t_0)$, and $f \in C^\infty(p)$, so that the velocity vector to the curve at p is defined by

$$(14) \quad F_*(\partial_{t,t_0}) f = \partial_t (f \circ F)_{t_0}.$$

Given a chart $(U \ni p, x^1, \dots, x^n)$, so that in local coordinates $\hat{F} = (f^1(t), \dots, f^n(t)) \triangleq (x^1 \circ F(t), \dots, x^n \circ F(t))$. Then, we may extract the i -th component of the velocity vector as $F_*(\partial_{t,t_0})x^i = \partial_t(x^i \circ F)_{t_0} = \dot{f}^i$, with the dot indicating the derivative with respect to t (often interpreted as time). The velocity vector may then be expanded in the basis of $T_p N$ as $F_*(\partial_{t,t_0}) = \dot{f}^i(t_0)E_{ip}$.

2.4.6. Generalizing to arbitrary parameterizations. The setup in the previous section discussed a parameterization and tangent space (spanned by the velocity vector) of a one dimensional submanifold of N . This is a special case of arbitrary (finite) dimensional parameterizations, which we describe next.

Consider an n -dimensional manifold $N \subset \mathbb{R}^m$, with $n < m$, an open set $W \subset \mathbb{R}^n$, and a parameterization (imbedding) $\theta : W \rightarrow N$ of a neighbourhood $V = \theta(W)$ in N . In fact, (V, θ^{-1}) is a chart on N . Let $u^1, \dots, u^n$ be the natural (global) coordinates in $\mathbb{R}^n$ (and therefore also on W) and let $x^1, \dots, x^m$ be the natural coordinates in $\mathbb{R}^m$ (and therefore also V , despite it only being n -dimensional). The (global) coordinate frames are $\{\partial/\partial u^1, \dots, \partial/\partial u^n\} \subset T\mathbb{R}^n$ and $\{\partial/\partial x^1, \dots, \partial/\partial x^m\} \subset T\mathbb{R}^m$. The local representative of θ may

be written in components as $x^j = \theta^j(u^1, \dots, u^n)$ for $j = 1, \dots, m$, and we know from Theorem 2.6 that at a point $\theta(u_0) \in V$, we have

$$(15) \quad (X_{u^i})_0 \triangleq \theta_* \left(\frac{\partial}{\partial u^i} \Big|_{u_0} \right) = \left(\frac{\partial \theta^j}{\partial u^i} \right)_{u_0} \frac{\partial}{\partial x^j} \Big|_{\theta(u_0)}.$$

Due to the linearity of θ_* , we have

$$(16) \quad \begin{aligned} \theta_* \left(\alpha^i \frac{\partial}{\partial u^i} \Big|_{u_0} \right) &= \alpha^i \left(\frac{\partial \theta^j}{\partial u^i} \right)_{u_0} \frac{\partial}{\partial x^j} \Big|_{\theta(u_0)} \\ &=: \beta^j \frac{\partial}{\partial x^j} \Big|_{\theta(u_0)} \end{aligned}$$

so that $\beta^j = \alpha^i (\partial \theta^j / \partial u^i)_{u_0}$.

Because θ is an imbedding, the Jacobian $(\partial \theta^j / \partial x^i)_{u_0}$ has rank n (see the definitions of immersion and imbedding in [Boothby(2002)] for details³), and thus the vectors $(X_{u^i})_0$, $i = 1, \dots, n$ span $T_{\theta(u_0)}N$, which is a subspace of $T_{\theta(u_0)}\mathbb{R}^m$.

An alternate description of the tangent plane at a point $\theta(u_0)$ is the collection of all tangent vectors at that point to curves passing through that point which lie in N . To give a little more detail, we proceed as follows.

Let $I = (a, b) \ni t_0$, and consider a curve $F = (f^1, \dots, f^n) : I \rightarrow W$, with $u_0 = F(t_0)$ and the corresponding curve on N given by

$$(17) \quad \theta \circ F(t) = (\theta^1(f^1(t), \dots, f^n(t)), \dots, \theta^m(f^1(t), \dots, f^n(t)))$$

The pushforward of ∂_t via this curve (with things evaluated at t_0 , $F(t_0)$ and $\theta(F(t_0))$) is

$$(18) \quad \begin{aligned} (\theta \circ F)_* \partial_t &= \partial_t (\theta^j(f^1(t), \dots, f^n(t))) \frac{\partial}{\partial x^j} \\ &= \frac{\partial \theta^j}{\partial u^i} \frac{df^i}{dt} \frac{\partial}{\partial x^j} \\ &= \dot{f}^i \left(\frac{\partial \theta^j}{\partial u^i} \frac{\partial}{\partial x^j} \right) \\ &= \dot{f}^i X_{u^i}, \quad \text{by Eq. 15.} \end{aligned}$$

Consider the coordinate curve, obtained by setting (along the first coordinate, say, without losing generality) $f^1(t) = t$ and $f^j(t) = u_0^j$ for $j = 2, \dots, n$. Then, Eq. 18 reduces to $(\theta \circ F)_* \partial_t = X_{u^1}$. Now observe that the $(X_{u^j})_0$ are the coordinate frame vectors (in $T_{\theta(F(t_0))}N$) corresponding to the chart (V, θ^{-1}) on N . Thus, the coordinate frame vectors are tangent to the coordinate curves.

2.5. Vector fields, flows, and the Lie derivative.

Definition 2.7 (Vector Field). A smooth vector field on a smooth manifold N is defined as a smooth map $X : N \rightarrow TN$ satisfying $\pi \circ X = i_N$, where i_N is the identity on N .

A vector field $X : N \rightarrow TN$ assigns a vector to each point on a manifold, such that the vector lies in the tangent space at that point, and the assignment is *smooth*. Smoothness is defined as follows.

For a chart $(U \ni p, \varphi)$ of N , and $\{E_{ip}\}_{i=1}^n$ a basis of $T_p N$, a vector may be written $X_p = \alpha^i(p) E_{ip}$. If p is varied in U , so that we consider different vectors in the vector field, the components α^i are real-valued functions on N , which we require to be smooth. In local coordinates, we require

$$(19) \quad \alpha^i = \alpha^i(x^1, \dots, x^n), \quad i = 1, \dots, n, \quad \text{on } \varphi(U) \subset \mathbb{R}^n$$

to be smooth, denoted by $\alpha^i \in C^\infty(\varphi(U))$. Note that the coordinate frames $E_1, \dots, E_n$ on U are independent vector fields, and one refers to them as the *field of frames* on U .

³replace with ref to the imbedding / rank theorem section above if/when that becomes available.

Remark 2.8. In Definition 2.7, the requirement

$$(20) \quad \pi \circ X = i_N$$

is there to ensure that a given vector in the vector field lies in the tangent space with the correct base point. At a more abstract level, a tangent bundle is an example of a *fiber bundle*, and a vector field is an example of a *smooth section* of this bundle, satisfying the property 20. Other examples of sections appear in differential geometry too, such as the dual notion of a one-form, which is a (smooth) section of the cotangent bundle (see Section 2.7).

A alternate coordinate-free definition of vector fields can be given by endowing TN with the structure of a smooth manifold, with $X : N \rightarrow TN$ a smooth mapping. In this scenario, a chart $(U, \varphi) = (U, x^1, \dots, x^n)$ on the manifold N induces a chart $(\pi^{-1}(U), \varphi_*)$ on the tangent bundle TN . For $X_p \in \pi^{-1}(U)$, expanded as $X_p = \alpha^i E_{ip}$, its local representative in $\varphi_*(\pi^{-1}(U))$ is written as $\hat{X}_a = \alpha^i \partial_{ia} = (x^1, \dots, x^n, \alpha^1, \dots, \alpha^n)$. One then argues that the natural projection $\pi : TN \rightarrow N$ is a C^∞ map between manifolds, and so is X , as a map from N into TN ; see [Nijmeijer and Van Der Schaft(1990)] and the exercises 5-7 in [Boothby(2002)] Section IV.2 for details.

2.5.1. Flow of a vector field.

Definition 2.9 (C^∞ action). The C^∞ action, $\theta : \mathbb{R} \times N \rightarrow N$ of $\mathbb{R}$ on a smooth manifold N is a C^∞ map between manifolds satisfying

$$(21) \quad \begin{aligned} \theta_0(p) &= p \quad \forall p \in N \\ \theta_s \circ \theta_t(p) &= \theta_{s+t}(p) = \theta_t \circ \theta_s(p) \quad \forall p \in N \text{ and } s, t \in \mathbb{R}. \end{aligned}$$

We note that this is an example of a one-parameter Lie group ($\mathbb{R}$) acting on a smooth manifold, where the group operation (+) is a smooth map, and the identity element is 0. The *orbit* of a point p is the set of all points “reachable” from this point, $\text{orb}_\theta(p) \triangleq \{\theta_t(p) \mid t \in \mathbb{R}\}$.

Definition 2.10. Given Definition 2.9, the *infinitesimal generator* of θ is defined as a vector field $X : N \rightarrow TN$ such that for each $p \in N$ and $f \in C^\infty(p)$,

$$(22) \quad X_p f = \lim_{\Delta t \rightarrow 0} \frac{f(\theta_{\Delta t}(p)) - f(p)}{\Delta t}$$

Theorem 2.11. The infinitesimal generator of an action is invariant under that action; that is, $\theta_{t*} X_p = X_{\theta_t(p)}$.

Proof. Consider $f \in C^\infty(\theta_t(p))$, and

$$(23) \quad \begin{aligned} \theta_{t*}(X_p)f &= X_p(f \circ \theta_t) \\ &= \lim_{\Delta t \rightarrow 0} \frac{f \circ \theta_t(\theta_{\Delta t}(p)) - f \circ \theta_t(p)}{\Delta t} \\ &= \lim_{\Delta t \rightarrow 0} \frac{f \circ \theta_{\Delta t}(\theta_t(p)) - f(\theta_t(p))}{\Delta t} \\ &= X_{\theta_t(p)}f, \end{aligned}$$

where we have used the fact that $\mathbb{R}$ is an Abelian group to swap the order of Δt and t . □

Corollary 2.12. If $X_p = 0$, then for each $q \in \text{orb}_\theta(p)$, $X_q = 0$.

Proof. This follows from the fact that θ_t is a diffeomorphism (being a C^∞ action of a Lie group on a smooth manifold),⁴ which implies that θ_{t*} is an isomorphism of $T_p N$ onto $T_{\theta_t(p)} N$, so that $X_{\theta_t(p)} = \theta_{t*} X_p = 0$ if and only if $X_p = 0$. □

Theorem 2.13. Depending on whether or not $X_p = 0$, $\text{orb}_\theta(p)$ is either just $\{p\}$ or an immersion of $\mathbb{R}$ in N .

⁴See Section III.7 in [Boothby(2002)]

Proof. Parameterize $\text{orb}_\theta(p)$ using parameter t by writing $F(t) = \theta_t(p)$. Consider an arbitrary point $t_0 \in \mathbb{R}$, and the vector $\partial_{t,t_0} \triangleq \frac{d}{dt}\big|_{t_0}$, which is a basis of $T_{t_0}\mathbb{R}$. From the rank theorem, we know that F is an immersion of $\mathbb{R}$ into N if and only if $F_*(\partial_{t,t_0}) \neq 0$ for all $t_0 \in \mathbb{R}$ (i.e., the Jacobian describing F_* has a constant rank (of 1) at all points on $\mathbb{R}$).⁵ Now consider $f \in C^\infty(F(t_0))$, and by the same arguments used in the proof of Theorem 2.11, we have

$$\begin{aligned} (24) \quad F_*(\partial_{t,t_0})f &= \partial_t(f \circ F)_{t_0} = \lim_{\Delta t \rightarrow 0} \frac{f \circ F(t_0 + \Delta t) - f \circ F(t_0)}{\Delta t} \\ &= \lim_{\Delta t \rightarrow 0} \frac{f \circ \theta_{\Delta t}(\theta_{t_0}(p)) - f(\theta_{t_0}(p))}{\Delta t} \\ &= X_{\theta_{t_0}(p)}f. \end{aligned}$$

Now, if $X_q \neq 0$ for any $q \in \text{orb}_\theta(p)$, then by Corollary 2.12, $X_{\theta_{t_0}(p)}$ is nonzero at all t_0 , and F is indeed an immersion. Furthermore, by the same corollary, if $X_{\theta_{t_0}(p)} = 0$, then this is true for all points on the orbit, and so F is a constant map (its Jacobian having a constant rank of 0).⁶ $\square$

Note that the formula just obtained, namely $F_*(\partial_{t,t_0}) = X_{F(t_0)}$ says that, given an action θ and its infinitesimal generator X , at each point p , X_p is the tangent velocity vector (see Section 2.4.5) of the curve $t \rightarrow F(t)$, that is, to the curve describing the orbit generated by this action. **We will often write $F_*(\partial_{t,t_0})$ as $\dot{F}(t_0)$ or $\dot{\theta}(t_0, p)$. Note that these, being tangent vectors, are different from the time derivatives of scalar quantities, $\dot{f}^i(t)$, introduced in Section 2.4.5.**⁷

Definition 2.14 (Integral Curve). Given a vector field X on a manifold N , we shall say that a curve $t \rightarrow F(t)$ defined on an open interval $(a, b) \subset \mathbb{R}$ is an *integral curve* of X if $\dot{F}(t) = X_{F(t)}$ on (a, b) .

In other words, the discussion above says that the orbit of an action is the integral curve of the infinitesimal generator of that action, $\dot{\theta}(t, p) = X_{\theta(t, p)}$ (for a given p).

One may define the derivative of various types of objects with respect to a vector field. The simplest of such objects are C^∞ functions, and their Lie derivative along a vector field coincides with the directional derivative defined earlier.

Definition 2.15 (Lie Derivative of a real-valued C^∞ function). Given a real-valued function $f : N \rightarrow \mathbb{R}$, and a vector field $X : N \rightarrow TN$, we define the Lie derivative of f with respect to X as

$$(25) \quad L_X f(p) \triangleq X(f)(p)$$

$$\begin{aligned} (26) \quad L_X f(p) &= \frac{d}{dt}\bigg|_{t=0} (f \circ \phi(t, p)) \\ &= \lim_{h \rightarrow 0} \frac{f(\phi(h, p)) - f(p)}{h} \end{aligned}$$

It can be shown that $L_X h(p)$ is equivalent to $X(p)h$ (the vector $X(p)$ acting as a differential operator—or *derivation*—on h at point p), and motivates the alternate notation $L_X h(p) = X(h)(p)$

do the bit with d/dt being pushed forward by a curve, defining the derivative

Let $X : N \rightarrow TN$ be a vector field on a manifold N . Recall the definition of the flow generated by a vector field: given X , an initial point p in some neighborhood $U \subset N$, a maximal interval of existence (a, b) of solutions, and a value $t \in (a, b)$, the flow $\phi(t, \cdot) : U \rightarrow N$ generated by X maps $p \mapsto \phi(t, p) \in N$, and satisfies the set of differential equations $\partial_t \phi(t, p) = X(\phi(t, p))$.⁸

Given a chart (U, φ) around $p \in N$, we may write $X(p)$ in local coordinates as $(X_1(x), \dots, X_n(x))^T$, where $x = \varphi(p)$

, where a vector field X may be written (with slight abuse of notation) as

⁵oct 7 '24: need to write at least high level / brief notes on the rank of a mapping, immersion and imbedding.

⁶revisit after doing the immersion / rank thm sections.

⁷I dont like any of the alts to $F_*(\partial_{t,t_0})$. They lose the idea that we are looking at a vector. Write the rest of the document then decide.. it might be best to use the $\dot{F}(t)$ alt, despite me not liking it (it doesnt have the base point). But the dot does implicate the velocity-ness wrt time.. of the displacement / flow curve F .

⁸Try to improve this flow part via one of the textbooks

and the Lie derivative might be written

This local representation illuminates that $X(p)h$, and therefore $L_X h(p)$, defines a real-valued function $X(h) : N \rightarrow \mathbb{R}$. This in turn allows for recursively defined Lie derivatives, $L_{X_1} L_{X_2} \cdots L_{X_r} h$ for some set of vector fields $\{X_1, \dots, X_r\}$.

2.6. The Lie Bracket, distributions and Frobenius' theorem.

- lie bracket, distribution, involutivity
- frobenius thm

2.7. Differential one-forms and codistributions.

2.7.1. *Covectors.* Recall from linear algebra that given a finite dimensional vector space V , we may associate with it a dual vector space V^* , whose elements $\sigma : V \rightarrow \mathbb{R}$ are the linear functionals or *covectors* on V . We briefly summarize some of their key properties.

- Addition and scalar multiplication in V^* are defined by

$$(\sigma_1 + \sigma_2)(v) = \sigma_1(v) + \sigma_2(v), \quad (\alpha\sigma)(v) = \alpha(\sigma(v)),$$

where the addition and multiplication on the RHS are of course operations in $\mathbb{R}$.

- For V, W vector spaces with their respective duals V^* and W^* , the linear maps $F_* : V \rightarrow W$ and $F^* : W^* \rightarrow V^*$ are related by $(F^*\sigma)(v) = \sigma(F_*(v))$. Injectivity (surjectivity) of F_* implies surjectivity (injectivity) of F^* .
- A basis $\{E_1, \dots, E_n\} \subset V$ induces a basis $\{\omega^1, \dots, \omega^n\} \subset V^*$ satisfying $\omega^i(E_j) = \delta_j^i$. This implies that for a vector $v = v^i E_i$, the coefficients can be extracted using the basis covectors, as in $v^j = \omega^j(v)$.

2.7.2. *One-forms and the differential of a function.* Given a manifold N , and its tangent space at p , $T_p N$, its dual *cotangent space at p* is denoted by $T_p^* N$, and these can be collected via disjoint union into a cotangent bundle, denoted by $T^* N$ (c.f. Eq. 7). Here, a covector is a mapping $\sigma_p : T_p^* N \rightarrow \mathbb{R}$, and as in the case of generic vector and covector spaces (Section 2.7.1), a basis $E_{1p}, \dots, E_{np}$ of $T_p N$ induces a covector basis $\omega_p^1, \dots, \omega_p^n$ of $T_p^* N$. A covector σ_p can be expanded in this basis as $\sigma_p = \sigma_p(E_{ip})\omega_p^i$, where $\sigma_{ip} \triangleq \sigma_p(E_{ip})$ is the coefficient of its i -th component. Considering covectors as a functions of points on a manifold leads to a notion dual to vector fields, called one-forms.

Definition 2.16 (One-form). A *smooth one-form* $\sigma \in T^* N$ assigns a covector $\sigma_p \in T_p^* N$ to each point p such that for any chart (U, φ) with frames $E_1, \dots, E_n$, the real-valued functions $\sigma_i = \sigma(E_i)$ are smooth (C^∞) on U .

Note that analogously to a vector field case, a smooth one-form is a (smooth) section of the cotangent bundle, satisfying $\tilde{\pi} \circ \sigma = i_N$, where $\tilde{\pi} : T^* N \rightarrow N$ is the natural projection, and i_N the identity.

The field of frames $E_1, \dots, E_n$ induces (pointwise) a *field of (coordinate) coframes* $\omega^1, \dots, \omega^n$ on U . This allows us to expand a one-form as the linear combination $\sigma = \sigma_i \omega^i$, with σ_i as in Definition 2.16. Given a vector field X and a one-form σ on $U \subset N$, $\sigma(X)$ is a smooth (since X and σ are smooth) real-valued function on U , with $(\sigma(X))(p) = \sigma_p(X_p)$.

Let $f : N \rightarrow \mathbb{R}$ be a smooth function and $X = \alpha^i E_i$ a smooth vector field. This defines a one-form df called the *differential* of f via the prescription $df(X) = X(f) \in C^\infty(N)$, where its value at a point is

$$\begin{aligned} df(X)(p) &= X_p(f) \\ (27) \quad &= \alpha^i(p) E_{ip}(f) \\ &= \alpha^i(p) \partial_{ip} f \quad (\text{dropping the caret from } \hat{f}) \end{aligned}$$

On an open set $U \subset \mathbb{R}^n$ with coordinate functions $x^1, \dots, x^n$ and field of frames $\partial_1, \dots, \partial_n$, we can define a field of coframes $dx^1, \dots, dx^n$, which is dual in the sense that $dx^i(\partial_j) = \delta_j^i$. We may extract the i -th component of a vector field X in the natural basis as $dx^i(X)$, and write $X = dx^i(X)\partial_i$.

To derive an expression for df in local coordinates, that is, to find the coefficients β_i in $df = \beta_i dx^i$, note that $\partial_j f = df(\partial_j) = (\beta_i dx^i)(\partial_j) = \beta_j$ so that

$$(28) \quad \begin{aligned} df &= (\partial_i f) dx^i \\ &= \frac{\partial f}{\partial x^1} dx^1 + \cdots + \frac{\partial f}{\partial x^n} dx^n. \end{aligned}$$

Remark 2.17. It is customary to write expansions in terms of covector bases $dx^1, \dots, dx^n$ as row vectors, while expansions in terms of vector bases $\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}$ as column vectors. Looking at Eqs. 27 and 28, we can write

$$(29) \quad df = \frac{\partial f}{\partial x^1} dx^1 + \cdots + \frac{\partial f}{\partial x^n} dx^n = \begin{bmatrix} \frac{\partial f}{\partial x^1} & \cdots & \frac{\partial f}{\partial x^n} \end{bmatrix},$$

$$(30) \quad X = \alpha^1 \frac{\partial}{\partial x^1} + \cdots + \alpha^n \frac{\partial}{\partial x^n} = \begin{bmatrix} \alpha^1 \\ \vdots \\ \alpha^n \end{bmatrix}$$

This allows us to write Eq. 27 in matrix form,

$$(31) \quad df(X) = \begin{bmatrix} \frac{\partial f}{\partial x^1} & \cdots & \frac{\partial f}{\partial x^n} \end{bmatrix} \begin{bmatrix} \alpha^1 \\ \vdots \\ \alpha^n \end{bmatrix},$$

which should be compared with Eq. 13.

Any real-valued smooth function f defines a one-form $df \in T^*N$ via Equation 27. However, not every one form can be written as a differential df of a function f . Those that can, are called *exact*. A necessary condition for a one-form $\sigma = \sigma_i dx^i$ to be exact is that it be closed, with the defining characteristic

$$(32) \quad \partial_j \sigma_i = \partial_i \sigma_j, \quad (\text{closed one-form}),$$

for all $i, j = 1, \dots, n$. This can be seen by recalling the identity $\partial_i \partial_j f = \partial_j \partial_i f$. The converse is partially true, where Condition 32 is sufficient for the local existence of a function f such that $\sigma_i = \partial_i f$.

2.8. more stuff. go do flows and lie derivatives and come back to this. Note also the relationship with the Lie derivative of a function

$$(33) \quad df(X) = X(f) = L_X f,$$

which will be useful in describing a characterization of nonlinear observability in Section XX.

!here

At every point $p \in N$, the *differential* of f at p is an element of T_p^*N , defined via its action on vectors,

$$(34) \quad \begin{aligned} ds(p)(X_p) &\triangleq X_p(s)(p) \\ &= \left(\sum_{j=1}^n X_{p,j} \frac{\partial}{\partial x_j} \Big|_p \right) s(p) \\ &= \sum_{i=1}^n \frac{\partial s}{\partial x^i} \Big|_p X_{p,i}, \end{aligned}$$

where we note that vectors are differential operators on real valued smooth functions.

We can endow $ds(p)$ with a representation in terms of the basis of T_p^*N by considering its action on a basis of T_pN , which results⁹ in $ds(p) = \sum_{i=1}^n \frac{\partial s}{\partial x^i}(p) dx^i|_p$. Then, (34) can be written as¹⁰, $ds(p)(X_p) = \left(\sum_{i=1}^n \frac{\partial s}{\partial x^i}(p) dx^i \right) \left(\sum_{j=1}^n X_{p,j} \frac{\partial}{\partial x_j} \right)$.

⁹Consider $ds(p) \frac{\partial}{\partial x^i} = \frac{\partial}{\partial x^i} s(p)$

¹⁰Noting the linearity of dx^i and right-distributivity of the expression in the first parentheses.

The *cotangent bundle* of a manifold N is defined as $T^*N \triangleq \cup_{p \in N} T_p^*N$, and can be given a manifold structure. Then, a *smooth one-form* σ on a smooth manifold N is defined as a smooth map $\sigma : N \rightarrow T^*N$ satisfying

$$\pi \circ \sigma = \text{identity (on } N),$$

where $\pi : T^*N \rightarrow N$ is the natural projection. That is, a one-form is a smooth map assigning a cotangent vector to each point $p \in N$.

In local coordinates x , one forms may be expressed¹¹ as $\sigma(x) = \sum_{i=1}^n \sigma_i(x) dx^i|_x$. Being dual objects of vector fields, one-forms act on them as expected: $\sigma(X)(p) = \sigma(p)X(p) \in \mathbb{R}$, and define real-valued smooth functions on N , $\sigma(X) : N \rightarrow \mathbb{R}$. Any real-valued smooth function s defines a one-form $ds \in T^*N$ via (34). Note the relationship with the Lie derivative, $ds(X) = X(s) = L_X s$. Note also that not every one form can be written as ds for some real-valued smooth function s . However, those that can, are called *exact*.

2.9. Codistributions. Recall that a (smooth) distribution [add to that section!](#) assigns a subspace of the tangent space to each point on the manifold in a smooth manner. Similarly, we can define a dual notion, the smooth codistribution. A smooth codistribution assigns, to each point on the manifold, a subspace of the corresponding cotangent space in a smooth manner (to be made precise below). Just as distributions and accessibility algebras play a fundamental role in describing nonlinear controllability, codistributions and *observation spaces* play a similar role in describing nonlinear observability (and by extension, identifiability). We discuss these ideas in the following sections.

Definition 2.18 (Smooth Codistribution). Around any point p , let there exist a neighborhood U of p and a set of smooth one-forms $\sigma_i \in T_p^*N, i \in I$ (I possibly infinite), such that for each $q \in U$, $P(q) = \text{span}\{\sigma_i(q); i \in I\}$. Then P is called a *smooth codistribution* on N .

In what follows, codistribution will always mean smooth codistribution. A one-form belongs to $P(p)$ if $\sigma(p) \in P(p)$ for any $p \in N$, and a codistribution is constant dimensional if the dimension of $P(p)$ does not depend on p . If a codistribution is constant dimensional of dimension l , then around each point p , there exist l independent one-forms (called the *local generators*) $\sigma_1, \dots, \sigma_l$ such that $P(q) = \text{span}\{\sigma_1(q), \dots, \sigma_l(q)\}$, for q near p .

Next, we define the notions of the kernel and annihilator of a codistribution and distribution respectively. Let P and D be a codistribution and distribution on N , respectively. Then,

$$(35) \quad \begin{aligned} \ker P(p) &= \{X(p) \mid X \text{ is a vector field s.t. } \sigma(X)(p) = 0, \forall \sigma \in P\} \\ \text{ann} D(p) &= \{\sigma(p) \mid \sigma \text{ is a one-form s.t. } \sigma(X)(p) = 0, \forall X \in D\} \end{aligned}$$

If D and P are constant dimensional, then $D = \ker(\text{ann} D)$ and $P = \text{ann}(\ker P)$. If $\ker P$ is involutive, then we call P an involutive codistribution. If P is generated by exact one forms, then it is easily shown that it must be involutive.

3. NONLINEAR OBSERVABILITY

Consider the nonlinear system given by

$$(36) \quad \begin{aligned} \dot{x} &= f(x) + \sum_{j=1}^m g_j u_j, & u &= (u_1, \dots, u_m) \in U \subset \mathbb{R}^m, \\ y_i &= h_i(x), & i &= 1, \dots, p, \end{aligned}$$

where $h = (h_1, \dots, h_p)^T : N \rightarrow Y = \mathbb{R}^p$ and $y(t, x_0, u) = h(x(t, x_0, u))$

We define the notions of state indistinguishability, local observability, observation space and the observability codistribution. We can then state two versions of the nonlinear observability rank condition. These will allow us to talk about decomposing observability into observable and unobservable modes, analogously to the linear case. These results will be extended to include identifiability using a simple device: including system parameters as additional state variables with zero dynamics, and considering their observability.

¹¹With slight abuses of notation, like dropping the hat from the local representative $\hat{\sigma}$.

Definition 3.1 (Nonlinear observability). Two states $x^1, x_2 \in N$ are *indistinguishable* for system (36) if for every admissible input function u the output functions $t \mapsto y(t, x^1, u), t \geq 0$ and $t \mapsto y(t, x_2, u), t \geq 0$ are identical on their common domain of definition. The system is *observable* if the states being indistinguishable implies $x^1 = x_2$.

Definition 3.2 (Nonlinear local observability). For an open set $V \subset M$, we say that $x^1, x_2 \in V$ are *V-indistinguishable*, denoted $x^1 I^V x_2$ if for every admissible *constant* control u such that $x(t, x^1, u)$ and $x(t, x_2, u)$ remain in V for $t \leq T > 0$, the output functions $y(t, x^1, u)$ and $y(t, x_2, u)$ are equal on their common domain of definition. The system (36) is called *locally observable* at x_0 if there exists a neighborhood W of x_0 such that for every neighborhood $V \subset W$ of x_0 the relation $x_0 I^V x^1$ implies $x^1 = x_0$. A system is locally observable if it is locally observable for all $x_0 \in M$.

A system is locally observable if every state x_0 can be distinguished from its neighbors by using system trajectories remaining close to x_0 .

Definition 3.3 (Observation space). The *observation space* $\mathcal{O}$ of the system (36) is the linear space (over field $\mathbb{R}$) of real valued smooth functions on N containing $h_i, i = 1, \dots, p$ and all repeated Lie derivatives

$$(37) \quad L_{X_1} \dots L_{X_k} h_j, \quad j = 1, \dots, p, \quad k = 1, 2, \dots$$

with $X_i \in \{f, g_1, \dots, g_m\}, i \in \{1, \dots, k\}$.

Proposition 3.4. $\mathcal{O}$ is equivalent to the linear space of functions on N containing $h_1, \dots, h_p$ and all repeated Lie derivatives along system trajectories. These Lie derivatives can be written $L_{Z_1} L_{Z_2} \dots L_{Z_k} h_j$, with $j \in \{1, \dots, p\}$ and $k = 1, 2, \dots$, and $Z_i, i \in \{1, \dots, k\}$ of the form

$$(38) \quad Z_i(x) = f(x) + \sum_{j=1}^m g_j(x) u_j^i,$$

for some point $u^i \in U$.

Proof. The linearity properties of the Lie derivative of a function, $L_{X_1+X_2} H = L_{X_1} H + L_{X_2} H$ and $L_X(H_1 + H_2) = L_X H_1 + L_X H_2$, together with the fact that the Z_i are linear combinations of $f, g_1, \dots, g_m$, imply $L_{Z_1} L_{Z_2} \dots L_{Z_k} h_j \in \mathcal{O}$. Conversely, all vector fields $f, g_1, \dots, g_m$ can be written as linear combinations of vector fields of the form Z_i . To see this, note that $f = Z_i$ for $u^i = 0$ for any i , and defining $Z_j^+ = f + g_j$ and $Z_j^- = f - g_j$ (that is, using $u = \pm(0, \dots, 0, 1, 0, \dots, 0)$ in (38), where the 1 is at the j -th coordinate), we have $g_j = \frac{1}{2}(Z_j^+ - Z_j^-)$. $\square$

Proposition 3.4 shows that $\mathcal{O}$ comprises the output functions and all derivatives of the output functions along system trajectories. In the case of systems with no inputs (or constant inputs), we can construct $\mathcal{O}$ using y_j and all repeated time derivatives, $\dot{y}_j = L_f h_j(x)$, $\ddot{y}_j = L_f L_f h_j(x)$, and so on. Initially, we will develop our results in this *autonomous* system setting.

Next, we define the central construct for gauging observability of a system.

Definition 3.5 (Observability Codistribution). Given $\mathcal{O}$, the *observability codistribution* $d\mathcal{O}$ is defined as

$$(39) \quad d\mathcal{O}(q) = \text{span}\{dH(q) \mid H \in \mathcal{O}\}$$

Since $d\mathcal{O}$ is generated by exact one-forms, it is involutive¹².

Theorem 3.6. Consider the system 36 with $\dim N = n$. Assume that $\dim d\mathcal{O}(x_0) = n$. Then the system is locally observable at x_0 .

The

4. TODOS

- Initial document should only have observability and identifiability. modify the intro and title to reflect that. the one with full controllability if version 2, coming after calibration is done.
- ...

¹²Write out involutivity of distributions and codistributions, and the exact one form stuff, frobenius. Also, check what the involutivity of the observability codistribution is needed for.

REFERENCES

- [Abraham et al.(1988)Abraham, Marsden, and Ratiu] Ralph Abraham, Jerrold E. Marsden, and Tudor Ratiu. *Manifolds, Tensor Analysis, and Applications*, volume 75 of *Applied Mathematical Sciences*. Springer, New York, NY, 1988. ISBN 978-1-4612-6990-8 978-1-4612-1029-0. doi: 10.1007/978-1-4612-1029-0. URL <http://link.springer.com/10.1007/978-1-4612-1029-0>.
- [Boothby(2002)] William M. Boothby. *An Introduction to Differentiable Manifolds and Riemannian Geometry, Revised*. August 2002. ISBN 978-0-08-057475-2. URL <https://shop.elsevier.com/books/an-introduction-to-differentiable-manifolds-and-riemannian-geometry-revised/boothby/978-0-08-057475-2>.
- [Nijmeijer and Van Der Schaft(1990)] Henk Nijmeijer and Arjan Van Der Schaft. *Nonlinear Dynamical Control Systems*. Springer, New York, NY, 1990. ISBN 978-1-4757-2102-7 978-1-4757-2101-0. doi: 10.1007/978-1-4757-2101-0. URL <http://link.springer.com/10.1007/978-1-4757-2101-0>.